Curriculum Vitae

Santiago Coelho

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Education

Ph.D. in Biomedical Engineering 2015–Jan. 2020
University of Leeds (2018–2020) and University of Sheffield (2015–2018), UK. Ph.D. awarded January 2020. Advisor: Prof. Alejandro Frangi.

Bachelor in Electrical Engineering 2010–2015
Five-year program, Universidad Nacional de La Plata (UNLP), La Plata, Argentina. Engineering Faculty Salutatorian.

Research and Teaching Experience

Assistant Professor (Research) Aug. 2026–present
Center for Biomedical Imaging (CBI), Department of Radiology, NYU Grossman School of Medicine.

Postdoctoral Research Fellow Jan. 2020–July 2026
MRI Biophysics Group (PIs: Dmitry Novikov and Els Fieremans), Center for Biomedical Imaging (CBI), Department of Radiology, NYU Grossman School of Medicine.

Teaching Instructor 2021–2024
BMSC-GA 4428: Practical Magnetic Resonance Imaging II, Vilcek Institute of Graduate Biomedical Sciences, NYU Grossman School of Medicine. Teaching site: Department of Radiology, Center for Biomedical Imaging. Three 2-hour lectures per year, November–December.

Graduate Teaching Assistant 2017–2018
Mathematics in Imaging Sciences (M.Sc. course), Department of Electronic and Electrical Engineering, University of Sheffield.

Teaching Assistant 2012–2015
Numerical Analysis, Faculty of Engineering, Universidad Nacional de La Plata (FiUNLP).

Teaching Assistant 2014
Control and Servomechanisms, FiUNLP.

Teaching Assistant 2013
Electromagnetism, FiUNLP.

Teaching Assistant 2011
Calculus II, FiUNLP.

Honors and Awards

- **Junior Fellow of the ISMRM** (2025).
- **NIH K99/R00 Award** (K99EB036080-01), NIBIB, NIH; approximately $1M for 2024–2029.
- Magna Cum Laude Award, ISMRM Annual Meeting, work #0641 (2024).
- Summa Cum Laude Award and AMPC Top 100 selection, ISMRM Annual Meeting, work #0069 (2023).
- Best Presentation Award, ISMRM Diffusion Study Group Workshop (2022).
- Magna Cum Laude Award, ISMRM Annual Meeting, work #0514 (2022).
- Best Presentation Award, ISMRM Diffusion Study Group, ISMRM Annual Meeting (2022).
- Best Presentation Award, ISMRM White Matter Study Group, ISMRM Annual Meeting (2022).
- Summa Cum Laude Award, ISMRM Annual Meeting, work #0397 (2021).
- Best Presentation Award, ISMRM Diffusion Day Trainee Workshop (2021).
- Award to the Best Graduate of Electrical Engineering, UNLP, La Plata, Argentina (2015).
- Annual Award from the Professional Council of Mechanical and Electrical Engineering for the highest GPA (2015).
- J. V. Gonzalez Award, La Plata City Council, for achieving one of the highest GPAs in FiUNLP (2015).
- Best Graduate Award, FiUNLP, for achieving the highest GPA in an engineering specialty (2015).
- Luis A. Huergo Award, Buenos Aires College of Engineers, for the highest GPA in a UNLP engineering specialty (2015).

Leadership

- **Elected Trainee Representative**, ISMRM Diffusion Study Group (approximately 1,000 members), June 2023–May 2025.
- **Chair**, Tissue Microstructure Imaging Gordon Research Seminar (GRS), 2023.
  Held in conjunction with the namesake Gordon Research Conference, the seminar provided a forum for doctoral and postdoctoral researchers to present work, discuss emerging methods and pre-publication data, and build collaborations. It covered multimodal approaches to tissue imaging at the cellular level, including microscopy, ultrasound, and MRI.
- **Co-Chair**, ISMRM Diffusion Trainee Day, 2024.
  This 24-hour virtual workshop spanned three time zones and included educational lectures, scientific and poster presentations, and networking activities for early-career researchers.
- **Co-supervisor**, M.Sc. project, “Multiple approaches for Diffusion Tensor Imaging estimation”, Yixin Huang, M.Sc. in Medical Imaging Sciences, University of Sheffield (February–August 2017). Main advisor: Prof. Alejandro Frangi.

Service and Editorial Activities

- **Organizing Committee**, ISMRM Diffusion Study Group Workshop, “40 Years of Diffusion: Past, Present & Future Perspectives”, Kyoto, February 2025. Co-Chair of the Educational and Historical Perspective Programs.
- **Organizing Committee**, ISMRM Member-Initiated Session, “Pushing the Boundaries of Diffusion MRI Across Field Strengths & Gradient Performance” (MIS-10), Singapore, 2024.
- **Guest Editor**, virtual special issue of *Magnetic Resonance in Medicine* on Diffusion MR (October 2023).
- **Co-Managing Guest Editor**, special issue of *Imaging Neuroscience* on Diffusion MRI (February–November 2026).
- **Reviewer** for *NeuroImage*, *Magnetic Resonance in Medicine*, *Imaging Neuroscience*, *Communications Biology*, *Human Brain Mapping*, *Medical Image Analysis*, *NMR in Biomedicine*, *Scientific Reports*, and *American Journal of Neuroradiology*.

Patents

1. **Coelho S**, Fieremans E, Novikov DS. System, method and computer-accessible medium for determining rotational invariants of cumulant expansion from one or more acquisitions which can be minimal. Assignee: New York University. U.S. provisional application 63/333,856, filed 04/22/2022; PCT/US2023/019622, filed 04/24/2023 (WIPO publication WO2023205506A2, 10/26/2023). U.S. application 18/923,344, filed 10/22/2024; published as US20250208246A1 on 06/26/2025. U.S. application pending; no U.S. grant date.
2. **Coelho S**, Fieremans E, Novikov DS. System, method and computer-accessible medium for diffusion MRI without shells. Assignee: New York University. U.S. provisional application 63/412,181, filed 09/30/2022; PCT/US2023/034248, filed 09/30/2023 (WIPO publication WO2024073122A1). U.S. application 19/095,419, filed 03/31/2025. U.S. patent US12638533B2, granted 05/26/2026.

Peer-Reviewed Publications

1. Leppert IR, Benbachir A, Campbell JSW, **Coelho S**, Feizollah S, Nelson MC, Brais B, Cocozza S, Pike GB, La Piana R, Tardif CL. Disentangling pontine fiber geometry and microstructure in ARSACS using advanced MRI. NeuroImage: Clinical. 2026;52:104066. doi:10.1016/j.nicl.2026.104066.
2. **Coelho S**, Chen J, Szczepankiewicz F, Fieremans E, Novikov DS. Geometry of the cumulant series in diffusion MRI. Nature Communications. 2026;17:4220. doi:10.1038/s41467-026-70018-w.
3. Chung S, Shin SH, Alivar A, McGiffin JN, **Coelho S**, Rath JF, Fieremans E, Novikov DS, El Berkaoui A, Foo FY, Rashbaum IG, Amorapanth P, Flanagan SR, Lui YW. Linking Symptom Phenotypes to Patterns of White Matter Injury in Mild Traumatic Brain Injury: A Latent Class Analysis. American Journal of Neuroradiology. 2026;47(5):1400-1405. doi:10.3174/ajnr.A9086.
4. **Coelho S**, Lemberskiy G, Zhu A, Lee HH, Abad N, Foo TKF, Fieremans E, Novikov DS. What if each voxel were measured with a different diffusion protocol? Magnetic Resonance in Medicine. 2026;95(4):2277-2290. doi:10.1002/mrm.70144.
5. Chan KS, Ma Y, Lee H, Marques JP, Olesen J, **Coelho S**, Novikov DS, Jespersen SN, Huang SY, Lee HH. In vivo human neurite exchange time imaging at 500 mT/m diffusion gradients. Imaging Neuroscience. 2025;3:imag_a_00544. doi:10.1162/imag_a_00544.
6. Ades-Aron B, **Coelho S**, Lemberskiy G, Veraart J, Baete SH, Shepherd TM, Novikov DS, Fieremans E. Denoising improves cross-scanner and cross-protocol test-retest reproducibility of diffusion tensor and kurtosis imaging. Human Brain Mapping. 2025;46(4):e70142. doi:10.1002/hbm.70142.
7. Coronado-Leija R, Abdollahzadeh A, Lee HH, **Coelho S**, Ades-Aron B, Liao Y, Salo RA, Tohka J, Sierra A, Novikov DS, Fieremans E. Volume electron microscopy in injured rat brain validates white matter microstructure metrics from diffusion MRI. Imaging Neuroscience. 2024;2:imag-2-00212. doi:10.1162/imag_a_00212.
8. **Coelho S**, Liao Y, Szczepankiewicz F, Veraart J, Chung S, Lui YW, Novikov DS, Fieremans E. Assessment of precision and accuracy of brain white matter microstructure using combined diffusion MRI and relaxometry. Human Brain Mapping. 2024;45(9):e26725. doi:10.1002/hbm.26725.
9. Chung S, Bacon T, Rath JF, Alivar A, **Coelho S**, Amorapanth P, Fieremans E, Novikov DS, Flanagan SR, Bacon JH, Lui YW. Callosal Interhemispheric Communication in Mild Traumatic Brain Injury: A Mediation Analysis on White Matter Microstructure Effects. American Journal of Neuroradiology. 2024;45(6):788-794. doi:10.3174/ajnr.A8213.
10. Liao Y, **Coelho S**, Chen J, Ades-Aron B, Pang M, Stepanov V, Osorio R, Shepherd T, Lui YW, Novikov DS, Fieremans E. Mapping tissue microstructure of brain white matter in vivo in health and disease using diffusion MRI. Imaging Neuroscience. 2024;2:imag-2-00102. doi:10.1162/imag_a_00102.
11. **Coelho S**, Baete SH, Lemberskiy G, Ades-Aron B, Barrol G, Veraart J, Novikov DS, Fieremans E. Reproducibility of the standard model of diffusion in white matter on clinical MRI systems. NeuroImage. 2022;257:119290. doi:10.1016/j.neuroimage.2022.119290.
12. Wilson LA, Heraty L, Ashford BA, **Coelho S**, Frangi AF, Pozo JM, Ince PG, Highley JR. Tissue microarray (TMA) use in post mortem neuropathology. Journal of Neuroscience Methods. 2021;347:108963. doi:10.1016/j.jneumeth.2020.108963.
13. Mozumder M, Pozo JM, **Coelho S**, Frangi AF. Population-based Bayesian regularization for microstructural diffusion MRI with NODDIDA. Magnetic Resonance in Medicine. 2019;82(4):1553-1565. doi:10.1002/mrm.27831.
14. Mozumder M, Pozo JM, **Coelho S**, Costantini M, Simpson JE, Highley JR, Ince PG, Frangi AF. Quantitative histomorphometry of capillary microstructure in deep white matter. NeuroImage: Clinical. 2019;23:101839. doi:10.1016/j.nicl.2019.101839.
15. **Coelho S**, Pozo JM, Jespersen SN, Jones DK, Frangi AF. Resolving degeneracy in diffusion MRI biophysical model parameter estimation using double diffusion encoding. Magnetic Resonance in Medicine. 2019;82(1):395-410. doi:10.1002/mrm.27714.
16. **Coelho S**, Pozo JM, Costantini M, Highley JR, Mozumder M, Simpson JE, Ince PG, Frangi AF. Histological data of axons, astrocytes, and myelin in deep subcortical white matter populations. Data in Brief. 2019;23:103762. doi:10.1016/j.dib.2019.103762.
17. Waller R, Baxter L, Fillingham DJ, **Coelho S**, Pozo JM, Mozumder M, Frangi AF, Ince PG, Simpson JE, Highley JR. Iba-1-/CD68+ microglia are a prominent feature of age-associated deep subcortical white matter lesions. PLOS ONE. 2019;14(1):e0210888. doi:10.1371/journal.pone.0210888.
18. **Coelho S**, Pozo JM, Costantini M, Highley JR, Mozumder M, Simpson JE, Ince PG, Frangi AF. Local volume fraction distributions of axons, astrocytes, and myelin in deep subcortical white matter. NeuroImage. 2018;179:275-287. doi:10.1016/j.neuroimage.2018.06.040.

Book Chapters

1. **Santiago Coelho**, Yogesh Rathi, and Marco Reisert. “Model Estimation,” in *Probing Brain Microstructure with MRI*, Springer Nature’s Neuroscience Methods series. Edited by Sune Jespersen and Noam Shemesh; to be published in 2027. *In preparation.*
2. Els Fieremans, Kouhei Kamiya, and **Santiago Coelho**. “Standard Diffusion MRI,” in *Diffusion MRI User Guide: A Window to Biological Tissue*, Springer Nature’s Diagnostic Imaging Series. Edited by Mami Iima, Susie Huang, and Denis Le Bihan; to be published in 2026. *Accepted; in editorial production.*

Manuscripts in Preparation

1. Valentin Stepanov, **Santiago Coelho**, Benjamin Ades-Aron, Jenny Chen, Timothy Shepherd, Ilya Kister, Dmitry Novikov, and Els Fieremans. “Advanced quantitative diffusion MRI markers unveil changes in the normal-appearing white matter reflecting disability in multiple sclerosis.” *In preparation; planned for submission in 2026.*
2. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “Zero-Shell Imaging (ZSI): microstructure-informed undersampling of diffusion MRI.” *In preparation; planned for submission in 2026.*
3. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “How do we know we measure tissue parameters, not the prior?” *In preparation; planned for submission in 2026.*

Peer-Reviewed Conference Articles

1. **Coelho S**, Pozo JM, Jespersen SN, Frangi AF. Optimal experimental design for biophysical modelling in multidimensional diffusion MRI. In: Medical Image Computing and Computer Assisted Intervention - MICCAI 2019. Lecture Notes in Computer Science. 2019;11766:617-625. doi:10.1007/978-3-030-32248-9_69.

Peer-Reviewed Conference Abstracts

Delivered 50+ presentations at ISMRM meetings (2017–2025), including the following first- or senior-author oral presentations and power pitches:

1. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “When fiber bundles are not axially symmetric: 4-dimensional fiber distributions in the human brain” Annual Meeting of the ISMRM, 2026.
2. Valentina Mazzoli, Smita Rao, . . . , Dmitry S. Novikov, Els Fieremans, and **Santiago Coelho**. “Microstructure and muscle strength in older adults: a time-dependent diffusion study.” Annual Meeting of the ISMRM, 2025.
3. **Santiago Coelho**, Filip Szczepankiewicz, Els Fieremans, and Dmitry S. Novikov. “Cumulant Tensors From The Addition Of Angular Momenta: All Diffusion Invariants In One Abstract.” Annual Meeting of the ISMRM, 2024.
4. **Santiago Coelho**, Valentin Stepanov, et al. “Sticks Or No Sticks? White Matter Microstructure In Multiple Sclerosis From High-B Scaling.” Annual Meeting of the ISMRM, 2024.
5. **Santiago Coelho**, Gregory Lemberskiy, et al. “What If Every Voxel Was Measured With A Different Diffusion Protocol?” Annual Meeting of the ISMRM, 2023.
6. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “Zero-Shell Diffusion MRI: Focus On Microstructure By Decoupling Fiber Orientations.” Annual Meeting of the ISMRM, 2023.
7. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “Rotational Invariants Of The Cumulant Expansion (RICE).” Annual Meeting of the ISMRM, 2022.
8. **Santiago Coelho**, Filip Szczepankiewicz, Els Fieremans, and Dmitry S. Novikov. “Fast Acquisitions For Mean Kurtosis And Microscopic Fractional Anisotropy.” Annual Meeting of the ISMRM, 2022.
9. **Santiago Coelho**, Filip Szczepankiewicz, et al. “Reproducibility Of White Matter Microstructure Mapping With Diffusion-Relaxometry.” Annual Meeting of the ISMRM, 2022.
10. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “How do we know we measure tissue parameters, not the prior?” Annual Meeting of the ISMRM, 2021.
11. **Santiago Coelho**, et al. “Optimal experimental design in multidimensional diffusion MRI for parameter estimation of biophysical tissue models.” Annual Meeting of the ISMRM, 2020.

Invited Talks

- **i2i Workshop, CAI2R, NYU Langone Health**. "Dimension Hopping Between Q-Space and K-Space." *Invited talk*; New York, October 22, 2026.
- **ISMRM White Matter Study Group Workshop**. "White matter microstructure through the lens of water diffusion." *Educational bootcamp lecture*; Faculty of Medicine, La Timone Campus, Marseille, France, October 13, 2026.
- **What Can You Be With a PhD? Symposium**. "Choosing a Traditional or Non-Traditional Postdoc." *Invited panelist*; NYU Langone Health, New York, October 26, 2025.
- **BrainMap Seminars, Martinos Center, Massachusetts General Hospital and Harvard Medical School**. "Brain microstructure imaging with diffusion MRI." *Invited research seminar*; Boston, October 15, 2025.
- **Department of Radiology, Weill Cornell Medicine**. "Diffusion MRI and tissue microstructure imaging." *Invited research seminar*; New York, October 8, 2025.
- **Institute of Imaging Science, Vanderbilt University**. "Diffusion MRI and microstructure imaging." *Invited research seminar*; Nashville, September 19, 2025.
- **Department of Radiology, Columbia University**. "Diffusion MRI and microstructure imaging." *Invited research seminar*; New York, August 10, 2025.
- **BMEII Seminars and Lectures, Icahn School of Medicine at Mount Sinai**. "Diffusion neuroimaging." *Invited research seminar*; Hess Center, New York, May 7, 2025.
- **Diffusion Microclub Scientific Seminars**. "Geometry of the cumulant series." *Invited research seminar*; Virtual, October 10, 2024.
- **ISMRM Annual Meeting**. "Diffusion Microstructure Models in the Brain." *Educational lecture*; Toronto, Canada, June 3, 2023.
- **ISMRM Diffusion Study Group Virtual Workshop**. "Brain microstructure: biophysical models." *Educational lecture*; Virtual, December 8, 2021.
- **ISMRM Annual Meeting, Brain Microstructure course**. "Diffusion: Physical Models." *Educational lecture*; Virtual, May 16, 2021.
- **Faculty of Engineering, Universidad Nacional de La Plata**. "Diffusion MRI for Well-posed and Optimal White Matter Microstructure Characterisation: Beyond Single Diffusion Encoding." *Invited research seminar*; La Plata, Argentina, December 5, 2019.